

The Pervasive Blind Spot

Benchmarking VLM Inference Risks on Everyday Personal Videos

Shuning Zhang, Zhaoxin Li, Changxi Wen, Ying Ma, Simin Li,
Gengrui Zhang, Ziyi Zhang, Yibo Meng, Hantao Zhao, Xin Yi, Hewu Li



Beijing, China
16-18 October 2026

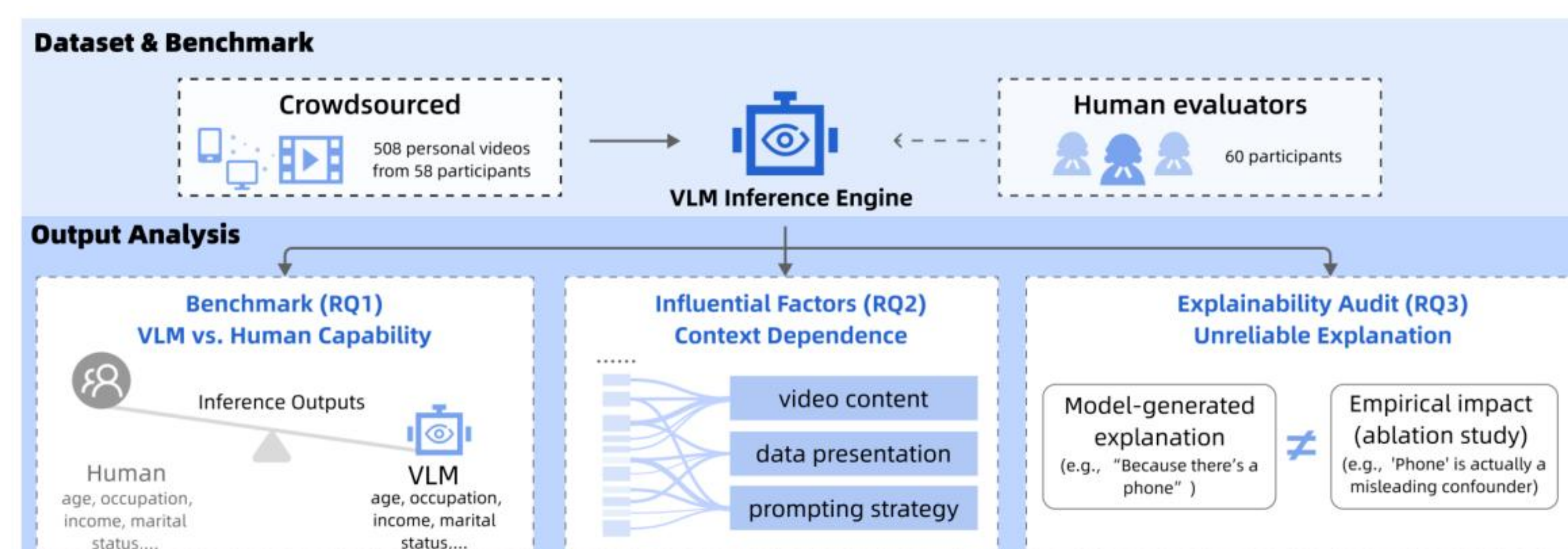


Everyday videos contain rich information, which tells a person's private attributes.

The Inference Issue

Cameras are everywhere. Smartphones, smart glasses and wearables turn daily life into a continuous visual stream, with much of it publicly shared.

VLMs change the threat. They *infer* what was never recorded: income, marital status, education, occupation, home district.



Dataset Distribution

508

videos from 58
creators
(filtered from 623 / 63)

642.8 min

total duration: 38,570 s
mean 75.9 s, median
25.6 s

Models Outperform Humans

Models perform better than humans on inference tasks.

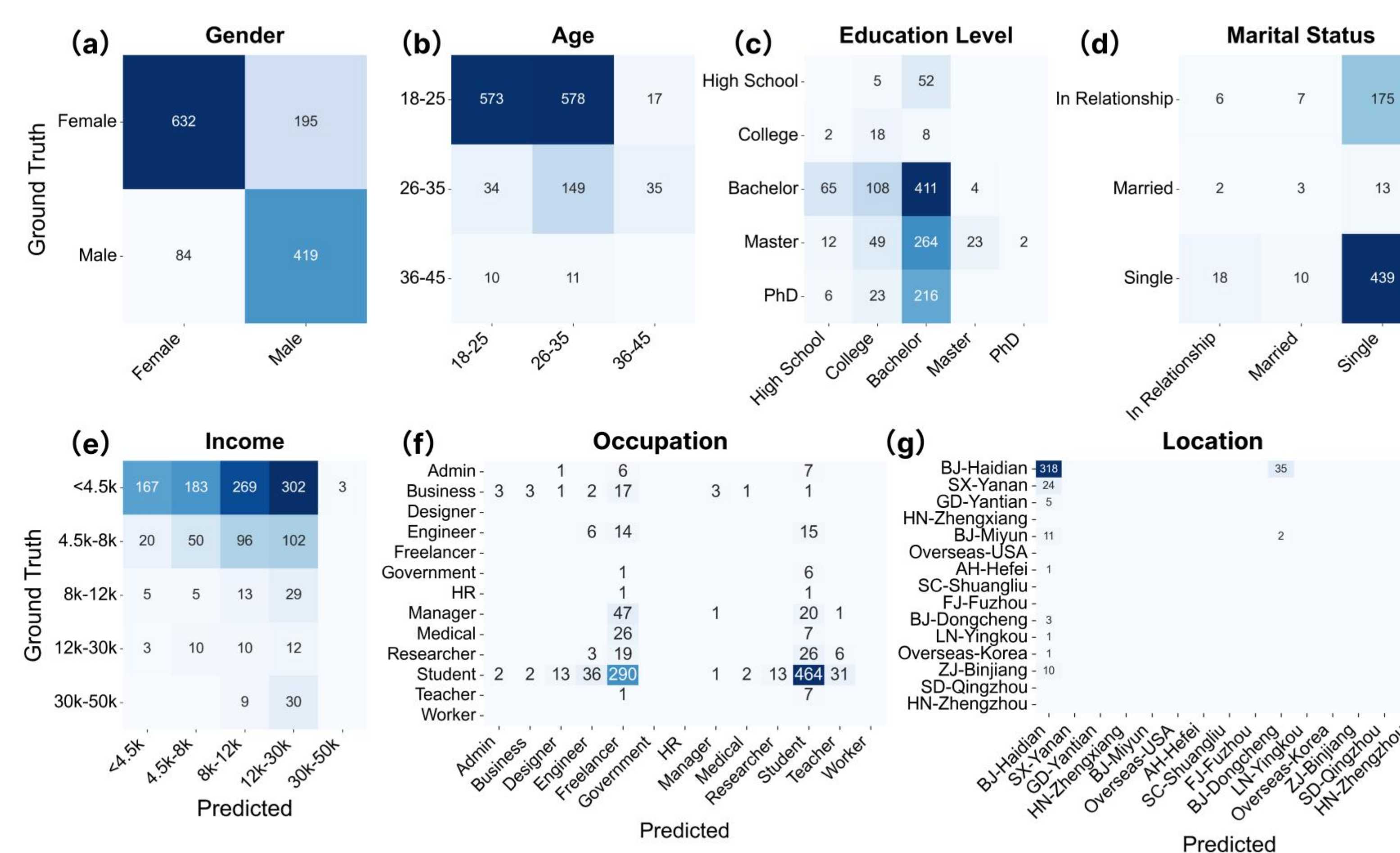
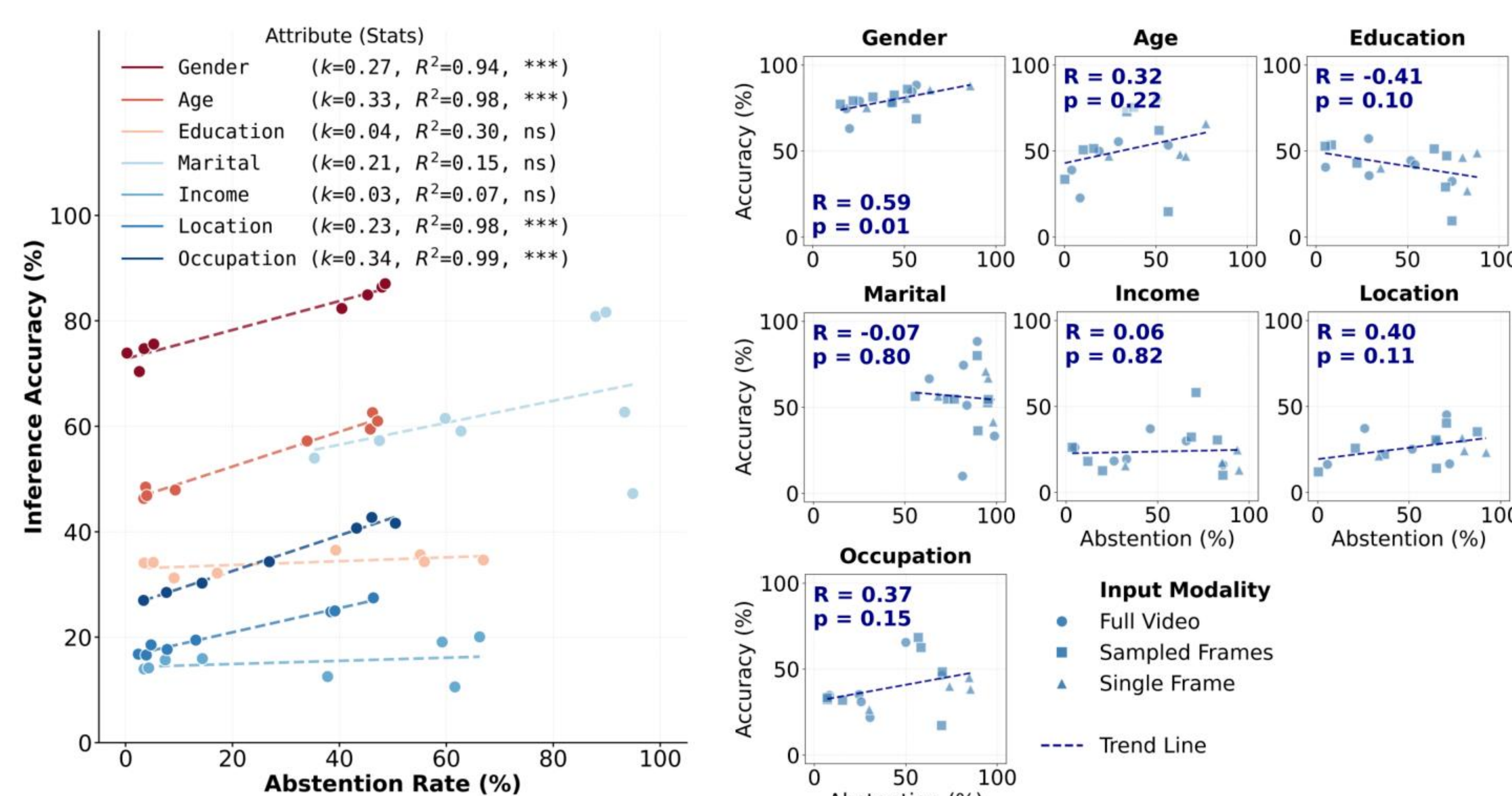
However, such advantage comes with abstention, where models could only answer better in a few cases.

Models Confidence Calibrate Poor

VLM's confidence for inference calibrates with their accuracy, but generalizes poorly across models.

Models Exhibit Certain Bias

Models bias towards certain "stereotypes" when conducting the inference, tending to classify attributes into certain classes.



Models' Explanation Aligns Poorly

Model's self-elicited explanation aligns poorly with the inference accuracy, where removing self-explained objects did not decrease inference accuracy.

Contact Us



My
Webpage



Contact



Paper

